Maison Test

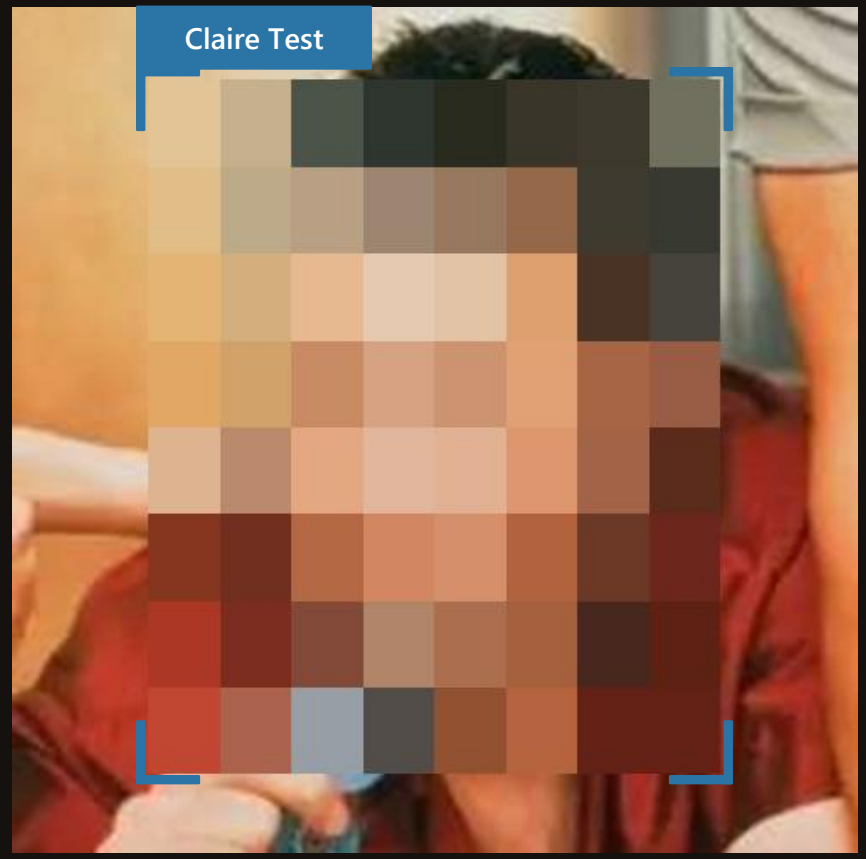
AUDENCIA · COURSE PROJECT · 2026

Recognizing a client in under a second

A client recognition prototype for a fictional luxury boutique, running entirely in the browser.

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Live demo mathos34.github.io/reconnaissance-equipe



Faces are pixelated in every screenshot.

THE IDEA

In a luxury boutique, the welcome happens in seconds

« Bonjour Madame Test, un café glacé sans sucre, comme d'habitude ? »

"Good morning Mrs Test, an iced coffee with no sugar, as usual?"
The welcome line the app writes from the client card and the time of day.

01 **Recognize without asking**

A registered client is identified as soon as they step into the camera's view.

02 **Remember preferences**

Drink, sugar, milk, style, recent purchases, advisor notes.

03 **Fit the moment**

The suggestion changes before and after noon and recalls the last visit.

Our question: how far can AI help a sales advisor, and where does it stop?

A complete app, published and usable from a link



1 Live recognition

A frame and a name on each face, “Inconnu” (unknown) otherwise.

2 Live indicators

Speed, delay and similarity, measured while it runs.

3 Client card

Preferences, purchases, advisor notes, next appointment.

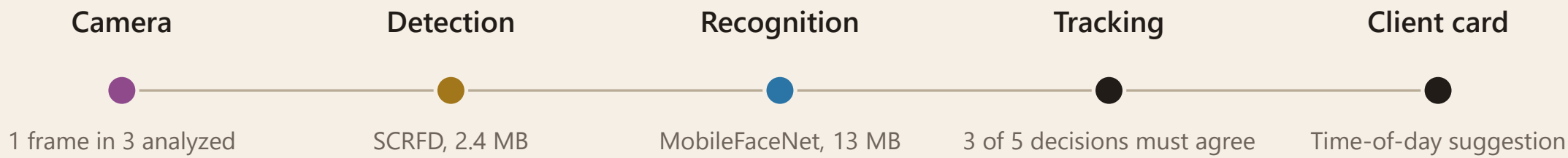
4 Welcome line

Written from the card and the time of day.

Two more pages: Clients (consent, guided capture, card) and Indicators (dashboard). The interface is in French.

HOW IT WORKS

No server: all the AI runs in the browser



0 images leave the device

Unknown faces are processed in memory and never stored.

15.4 MB of models

Downloaded once, then cached. ONNX Runtime Web, on WebGPU or WebAssembly.

512 numbers per client

The only biometric data kept, after typed consent. No photo is stored.

Same models as our first Python version (InsightFace buffalo_s), ported to JavaScript and checked against it: faces within 0.17 px, signatures at 0.9985 similarity.

RESULTS

The numbers, measured by the app itself

Median delay

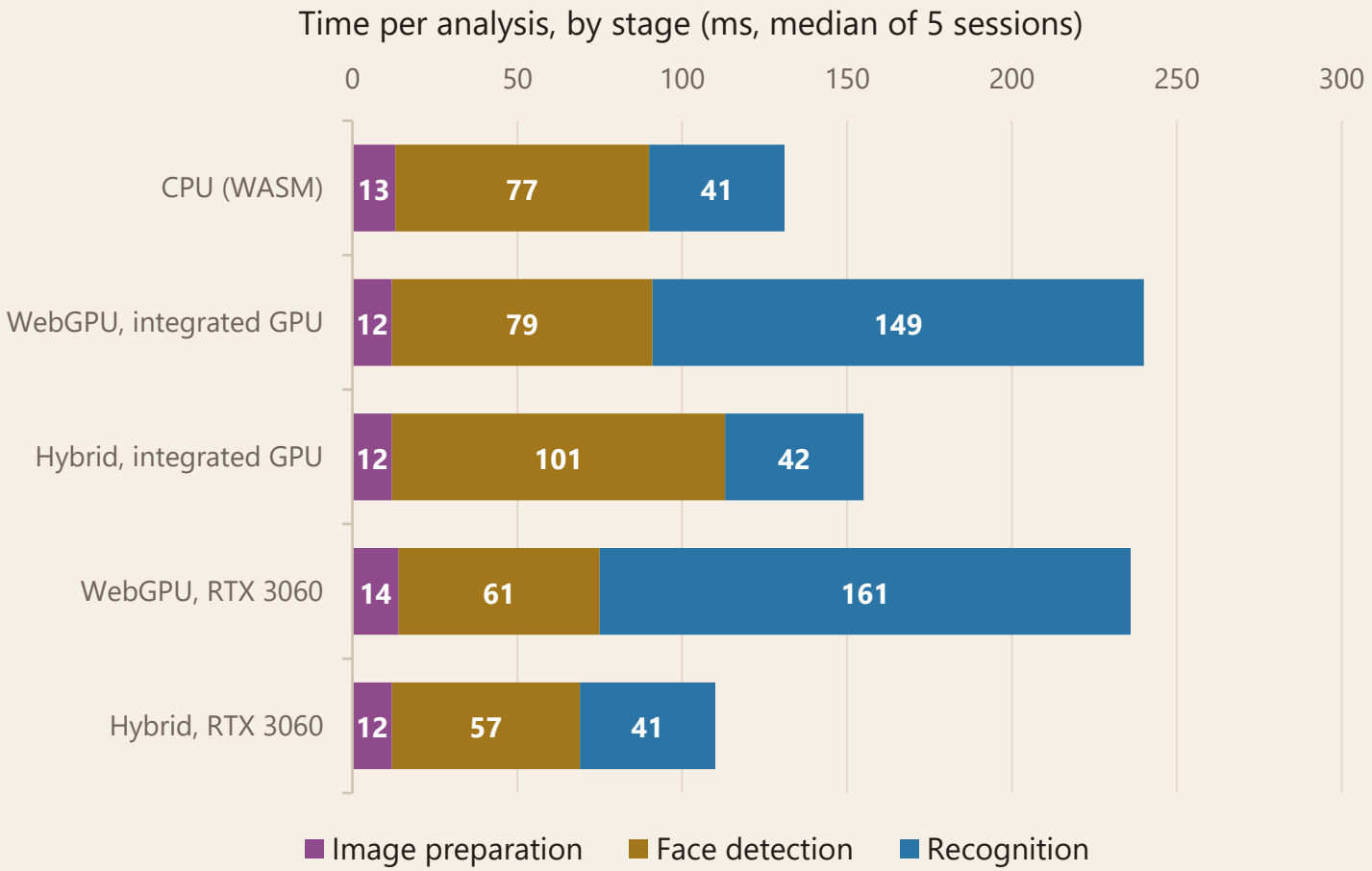
0.20 s

From a face entering the frame to the client card on screen, on the demo PC. 0.34 s on CPU only.

Time per analysis 132 ms on CPU only	108 ms
Display rate 8.1 analyses per second	24 fps
Enrolling a client 25 valid images across 5 poses	18.7 s
False accepts at threshold 0.35 6 identities, 7 degraded conditions	0%
Reliable range 83% still identified at 3.9 m	2.0 m
JavaScript vs Python gap 0.9985 signature similarity	0.17 px
Automated tests 45 web, 88 Python, plus end to end	133

Edge 156 · Ryzen 7 5800H · RTX 3060 · medians of 5 sessions of 12 s

Measuring changed our engine choice



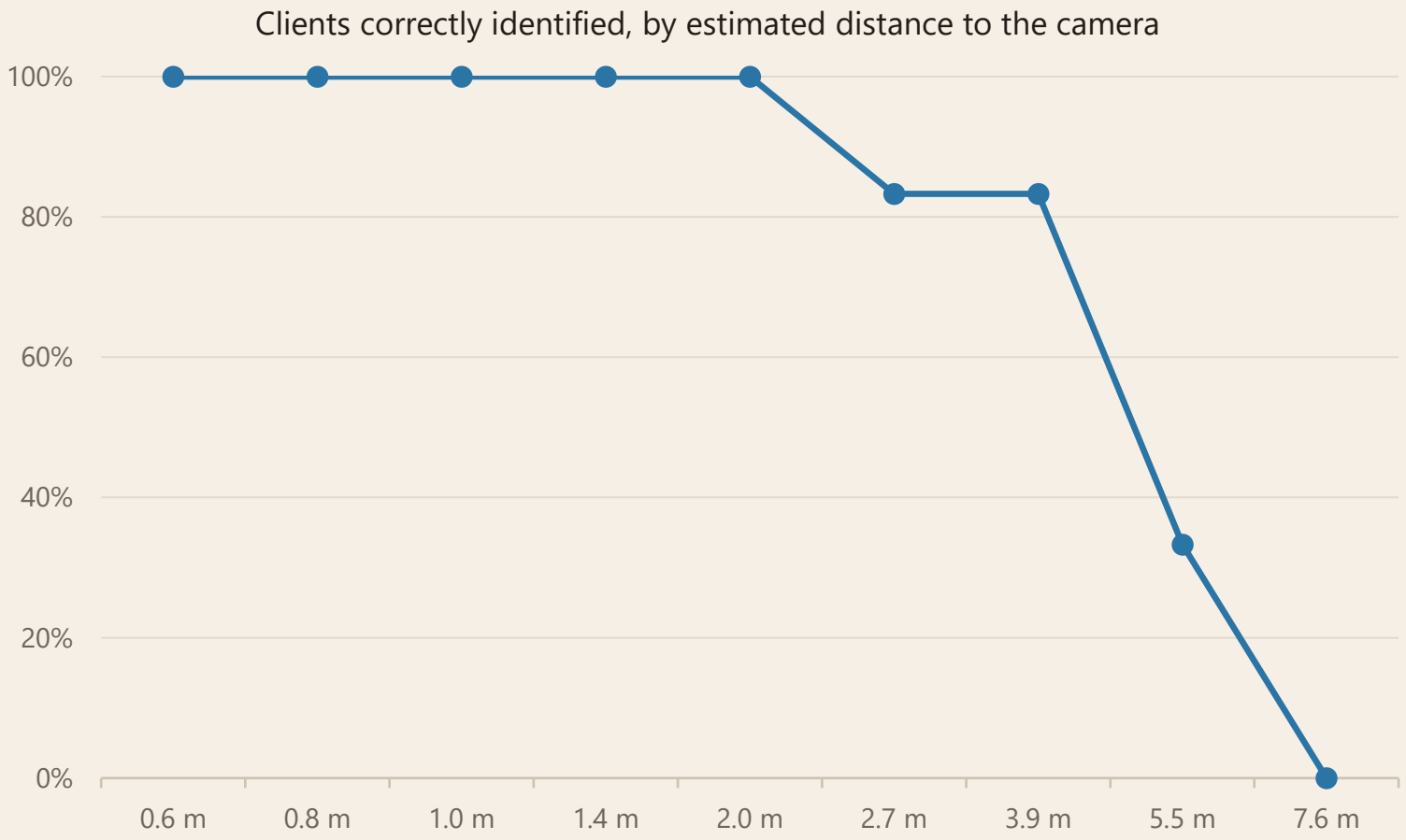
1. Recognition is 3 to 4 times slower on the graphics card, whichever card (149 to 161 ms vs 41 ms).

2. Detection only speeds up on the dedicated card (57 ms vs 77 ms).

3. So: CPU by default, hybrid on the demo PC. 108 ms per analysis, 0.20 s to the card.

Without measuring, we would have concluded "the GPU makes everything faster". The browser was in fact using the integrated GPU.

Reliable up to about 2 m from the camera

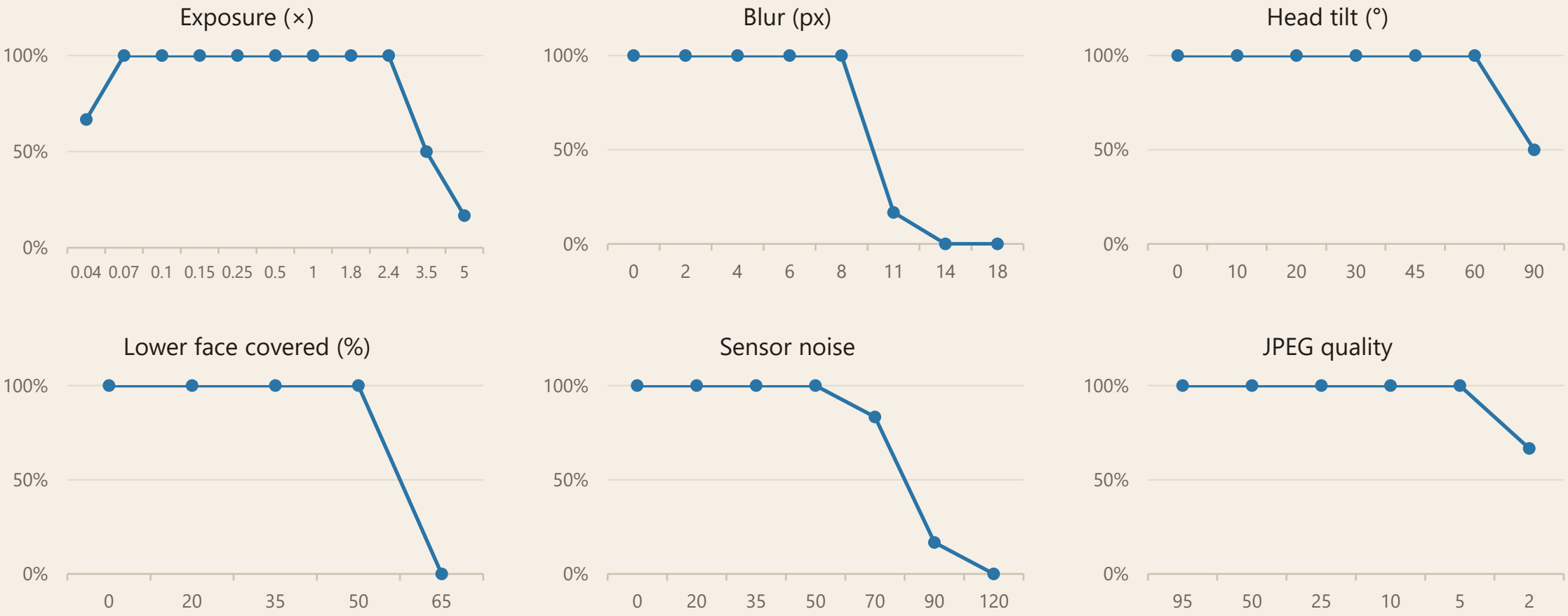


READING

1. 100% up to 2.0 m, still 83% at 3.9 m.
2. Beyond 4 m the face is under 35 px wide and is no longer detected.
3. Every detected face was recognized: detection is the limit.
4. Takeaway: camera near the entrance, at face height.

Distance estimated for an HD webcam with a 70° field of view and a 15 cm face. 6 identities, same source photo.

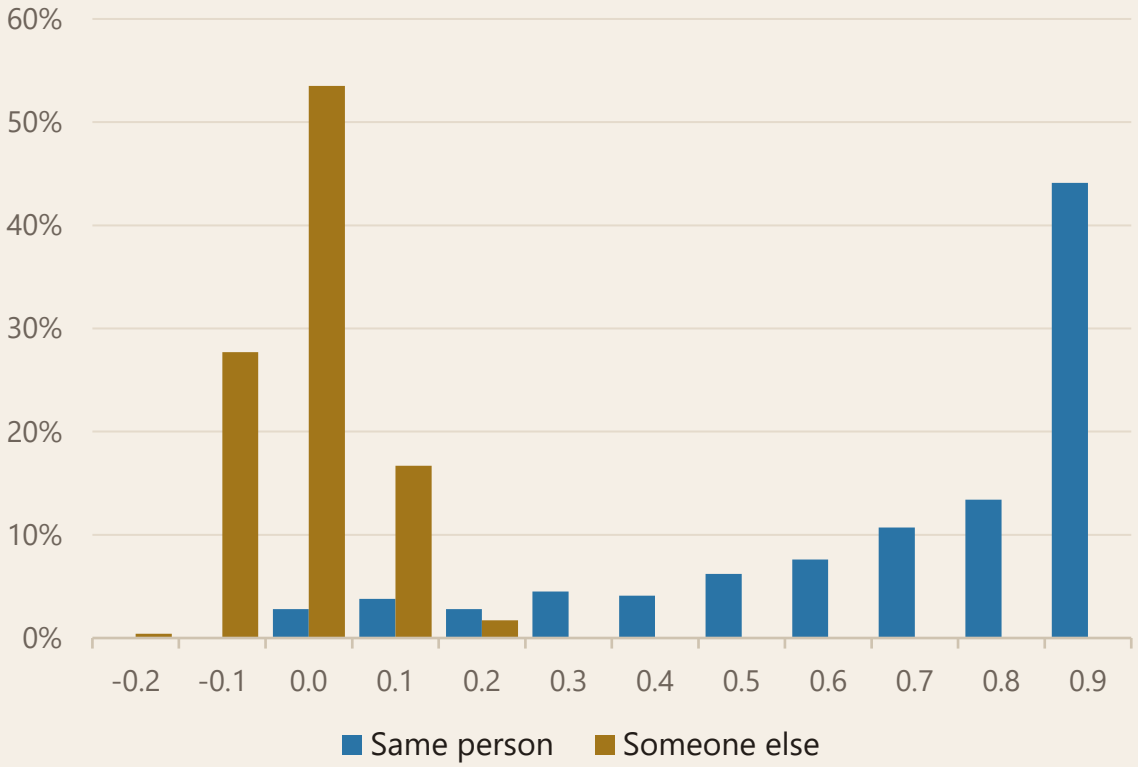
Handles light well, struggles with blur and covered faces



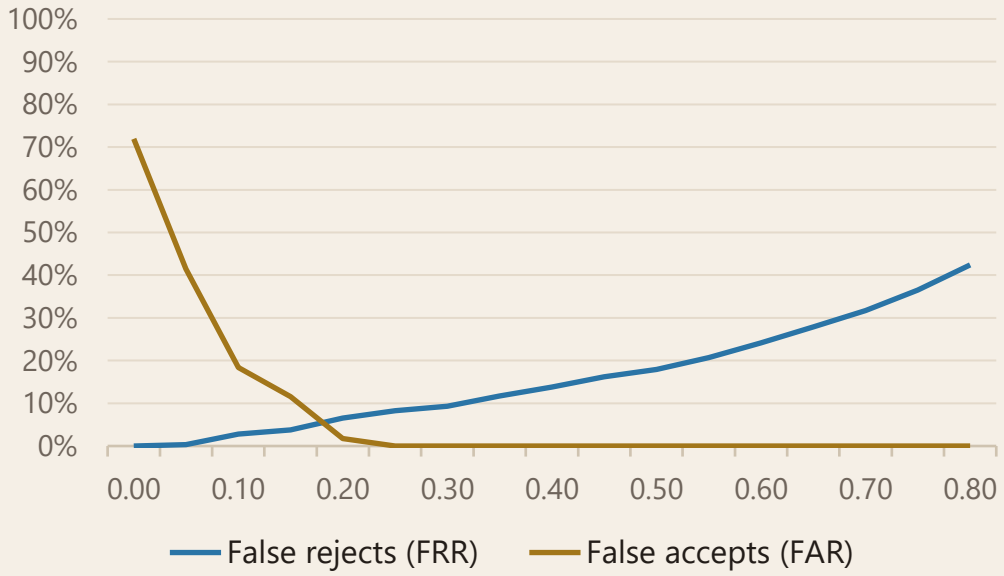
Correct identification rate at threshold 0.35, 6 identities, one point per level. Degradation increases to the right, except exposure: 1 = normal, darker to the left, brighter to the right.

At threshold 0.35: 0% false accepts, 11.7% false rejects

Face-to-template similarity (share of comparisons)



Error rates by threshold



The two groups are well apart: other people never go above 0.24 similarity. Threshold 0.35 favors never showing the wrong card: better an “Inconnu” than the wrong name.

Indicative evaluation: 6 identities, same source photo for enrollment and test, all degraded conditions combined.

LIVE DEMO

From sign-up to welcome in four minutes

0:00 — Sign up a client

Clients page: the person types J'ACCEPTE, then 25 images are captured across 5 poses.

1:15 — Fill in the card

Drink, sugar, milk, style, recent purchases, next appointment.

2:15 — Recognize them

Recognition page: the person steps in and the card appears. Another team member stays “Inconnu”.

3:15 — Read the indicators

Delay, time per stage, enrollment quality.

4:00 — Questions

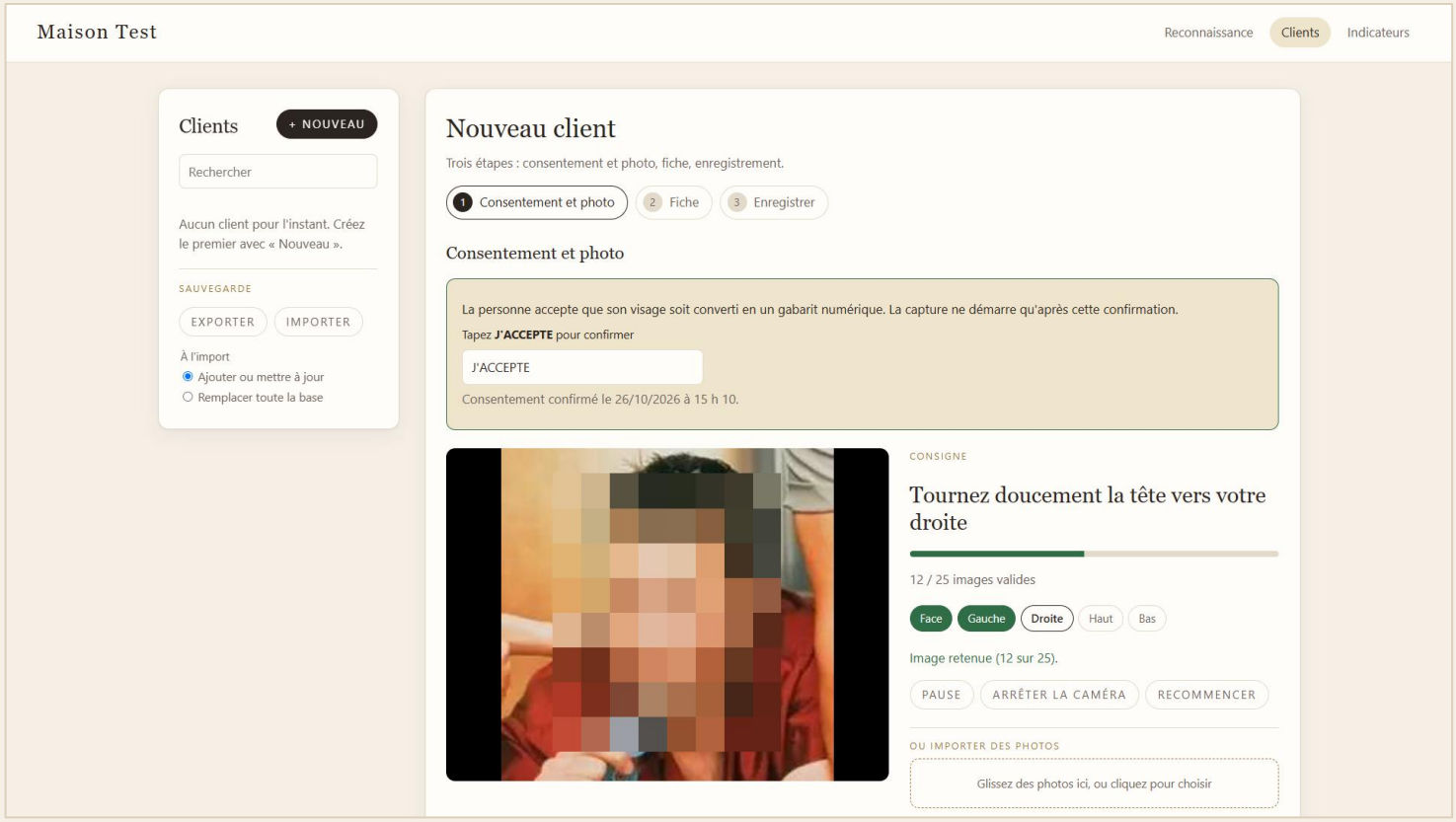
Try it

Open the demo on your phone while we run it.

PLAN B

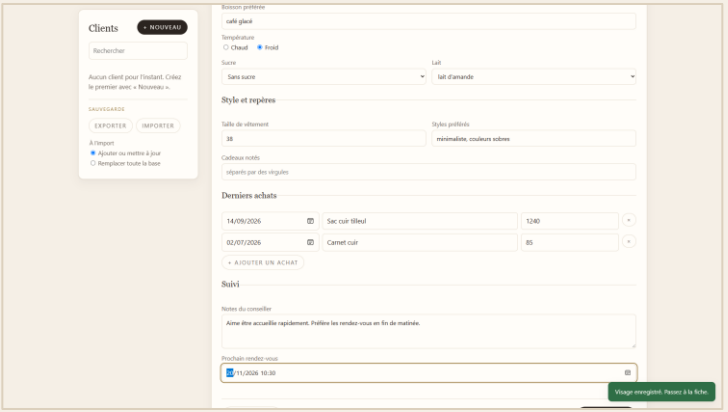
Open with ?moteur=wasm to use the CPU.
Import the prepared JSON backup.
Next slide: the same flow in screenshots.

The same flow in screenshots

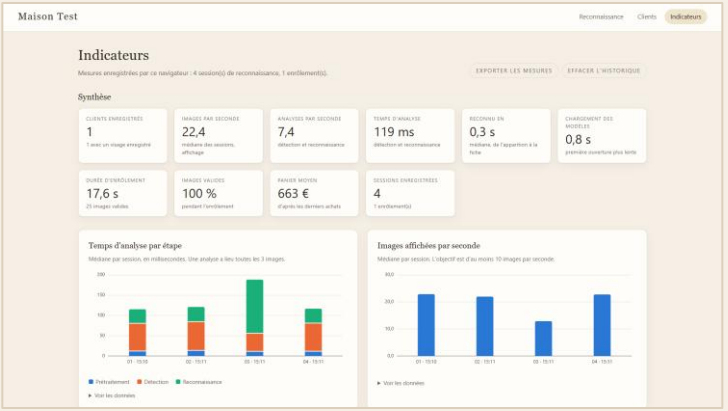


1. Consent first, then guided capture in 5 poses

Recognition itself is on slide 3. Simulated camera, faces pixelated.



2. Card: preferences, purchases, appointment



3. Indicators measured during the demo

Technical limits, and what we would do next

	WHAT WE SAW	WHAT WE WOULD DO
01	Evaluated on 6 identities, one photo per person	Test with a volunteer panel over several days, with glasses and different hairstyles
02	No spoofing detection: a photo of a photo works	Add liveness detection (blinking, depth)
03	Database stored in a single browser	Shared, encrypted database across the boutique's devices
04	Poses are guided but not checked during enrollment	Check head angle from the facial landmarks
05	132 ms per analysis, on a single CPU core	Enable multithreading (site headers) and an optimized model
06	Usable range about 2 m, sensitive to blur	Better placed, higher resolution camera, front lighting

Legally, biometrics is the real issue

REGULATION (EU) 2016/679

GDPR, Article 9

Biometric data is prohibited by default. The usable exception is explicit, freely given consent, so a face-free alternative must exist.

REGULATION (EU) 2024/1689

EU AI Act

Remote biometric identification is listed as high risk (Annex III). The obligations were pushed back to 2 December 2027 by the June 2026 Omnibus agreement.

CNIL, FRENCH DATA PROTECTION AUTHORITY

Consent and impact assessment

Walking into a shop is not consent. A data protection impact assessment is required before any deployment.

INSIGHTFACE

Model license

The models we used are for non-commercial research only. A real deployment needs commercially licensed models.

What we applied: dated "J'ACCEPTE" consent before any capture, no photo stored, unknown faces never saved, deletion on request. Not an exhaustive analysis; to be reviewed by a lawyer.

The camera matters as much as the model

Camera	HD webcam at least. Beyond 2 m or with motion blur, recognition drops.
Lighting	Copes with dim light, but backlight or strong side light hurts detection.
Placement	At the entrance, at face height, facing the flow. Several people at once: one card shown.
Computer	A recent PC: 132 ms per analysis on a Ryzen 7 5800H. An older PC or a tablet will be slower.
Browser and network	Recent Chrome or Edge. 15 MB to download on first launch, then everything runs locally.

Order of magnitude per entrance: a €50 to €150 webcam and an existing PC are enough for the prototype. No server, no subscription.

Commercially, technology is not the main obstacle

01 **Acceptability**

In luxury, discretion comes first. Being recognized without asking can feel like surveillance.

02 **Real consent**

Service must be just as good without face recognition, or consent is not freely given.

03 **Costly mistakes**

Showing another client's card exposes their purchases and notes. Rare, but serious.

04 **Integration**

The value is in the client card, so it has to plug into the existing CRM.

A realistic positioning

- An opt-in program for loyal clients who ask for it, never on by default.
- The advisor stays in control: a Hide card button, no automatic action.
- A face-free alternative: loyalty card or app.
- Success metric: the share of invited clients who opt in, and their satisfaction.

TAKEAWAYS

What this project says about AI today

Already good at

- Recognizing a known face in a fraction of a second, on an ordinary PC.
- Running without a server, in a plain browser, with 15 MB of models.
- Staying stable in dim light or with a compressed image.

Still hard

- Real conditions: distance, blur, covered or turned faces.
- Spoofing, fairness across profiles, evaluation at scale.
- The framework: consent, biometrics law, client acceptance.

SKILLS WE USED

Product framing · *On-device AI* · *Measurement and KPIs* · *Digital law* · *Software quality*

Maison Test

NEXT STEPS

It works. Now it needs a framework.

1. Test with a volunteer panel, in a real boutique, over several days.
2. Add spoofing detection and switch to commercially licensed models.
3. Run the impact assessment and design the client opt-in journey.

Try it



Thank you. Demo mathos34.github.io/reconnaissance-equipe · Code github.com/Mathos34/reconnaissance-equipe

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